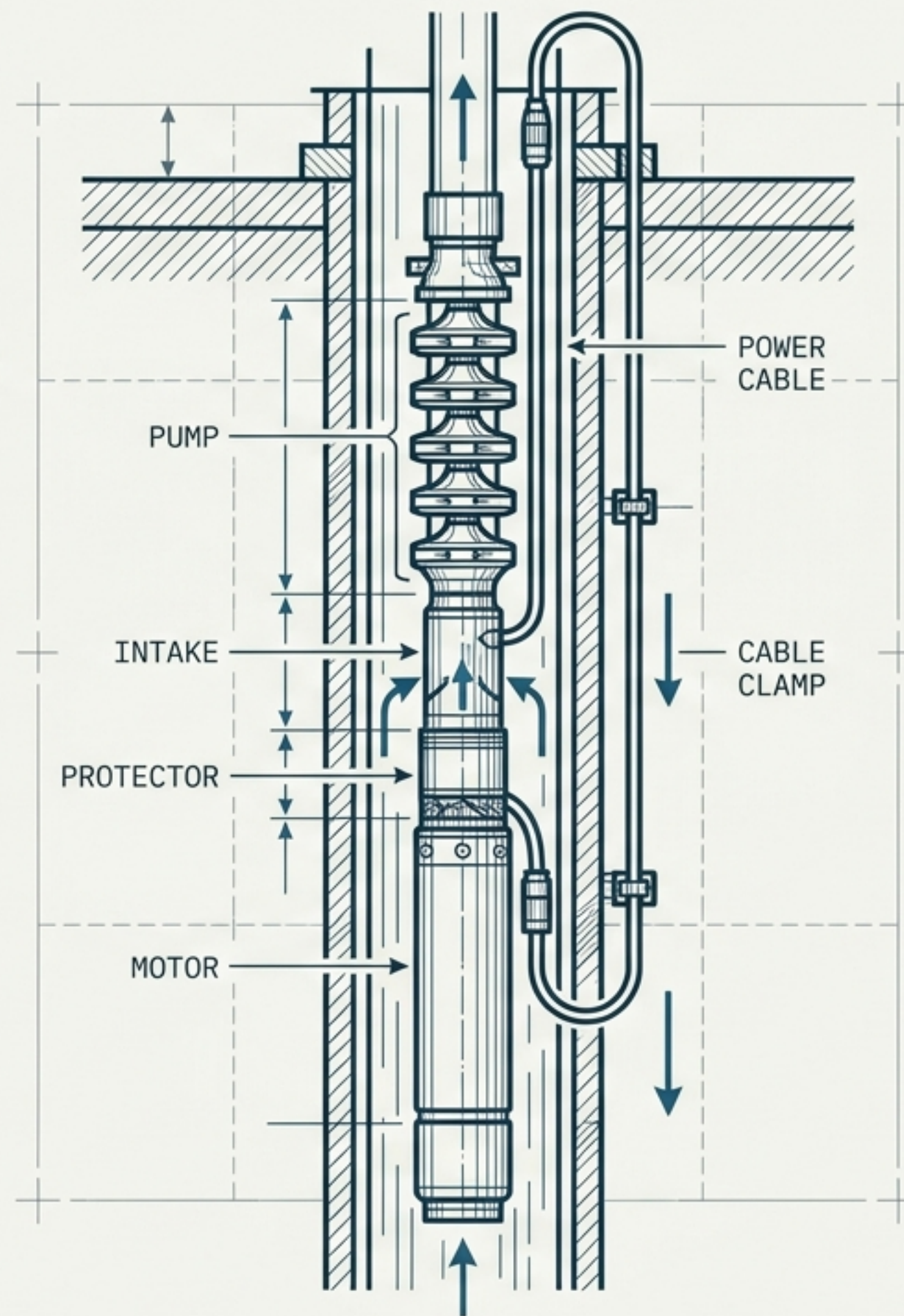


Electrical Submersible Pumps (ESPs): Core Engineering Principles

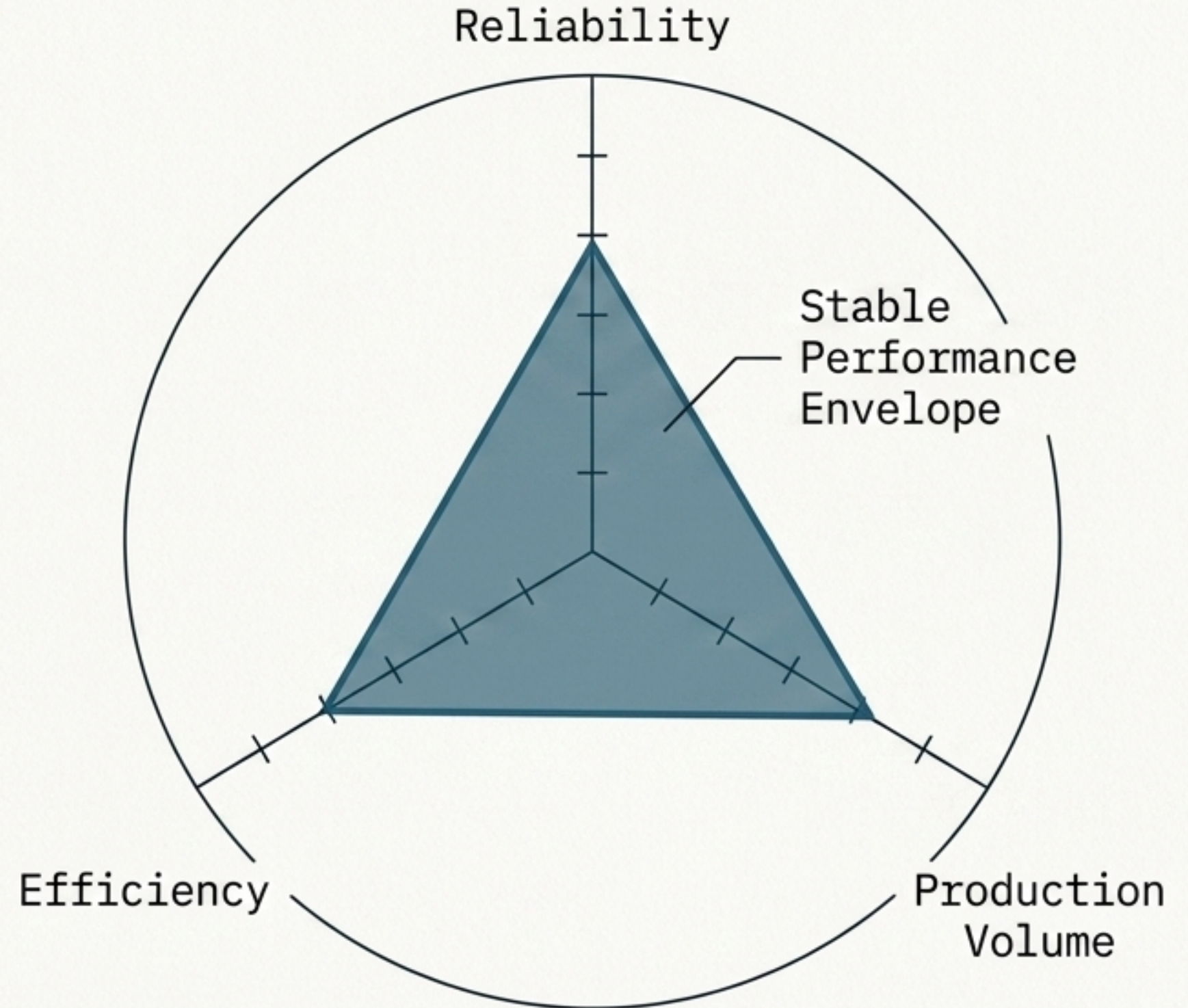
Authored by Engineer Sajjad K. Hano



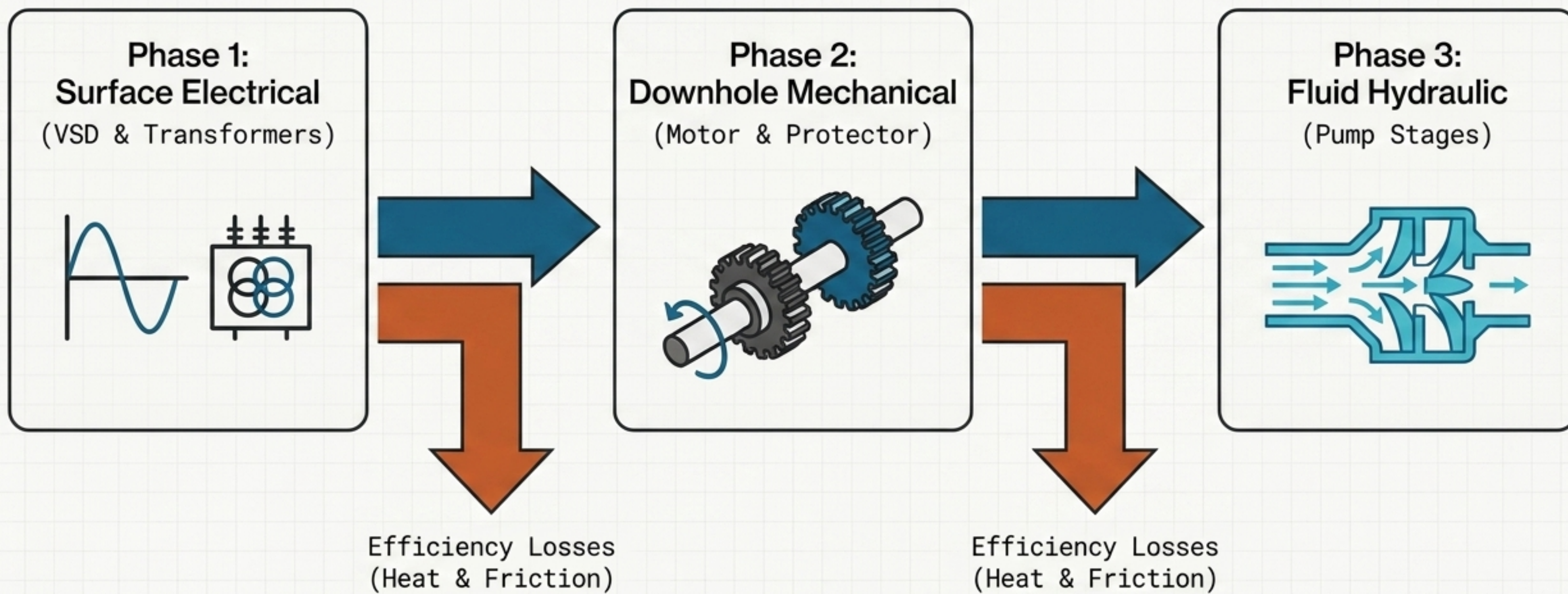
The Optimization Mandate

“Performance depends on much more than simply adding stages or increasing motor power.”

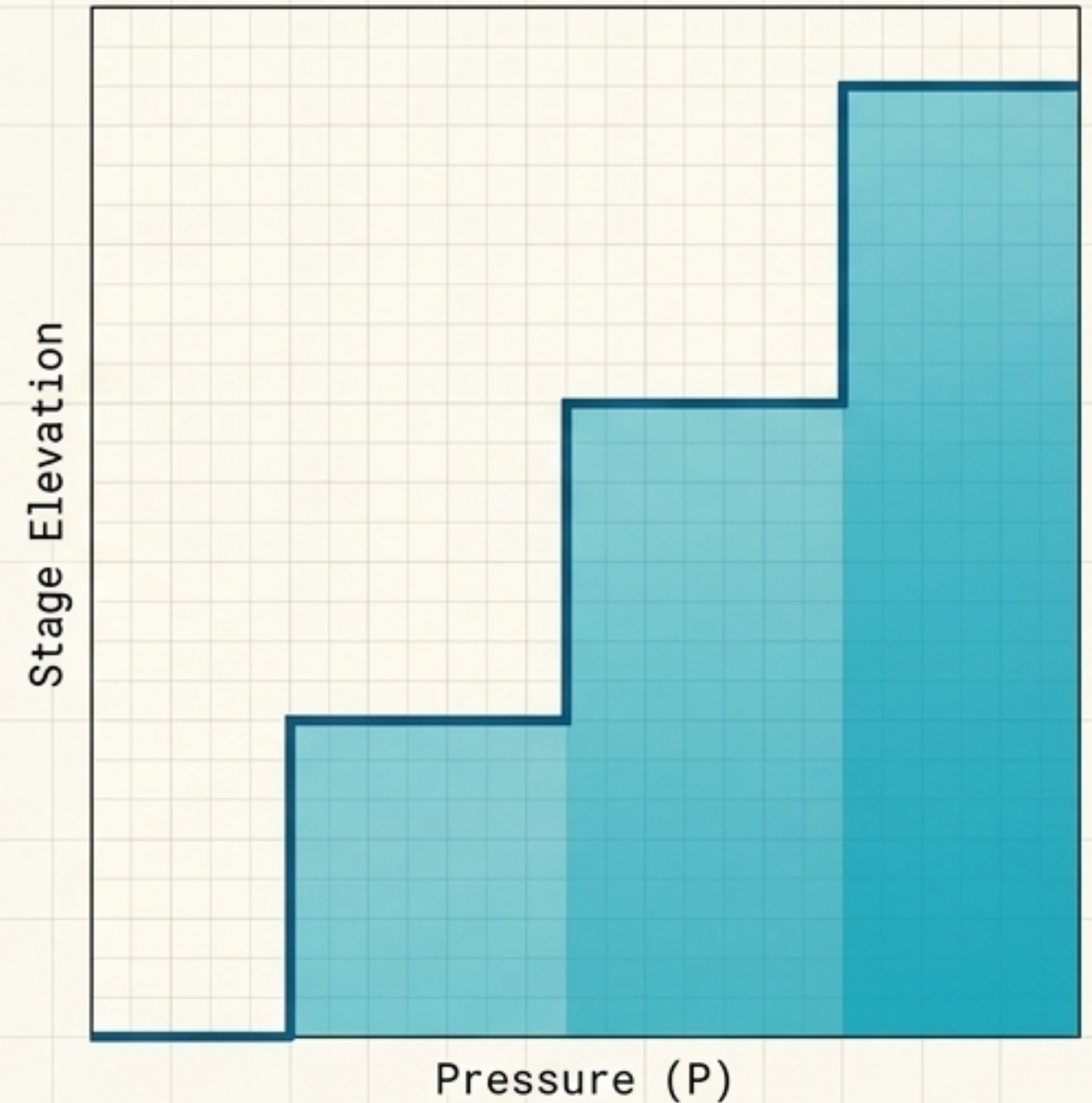
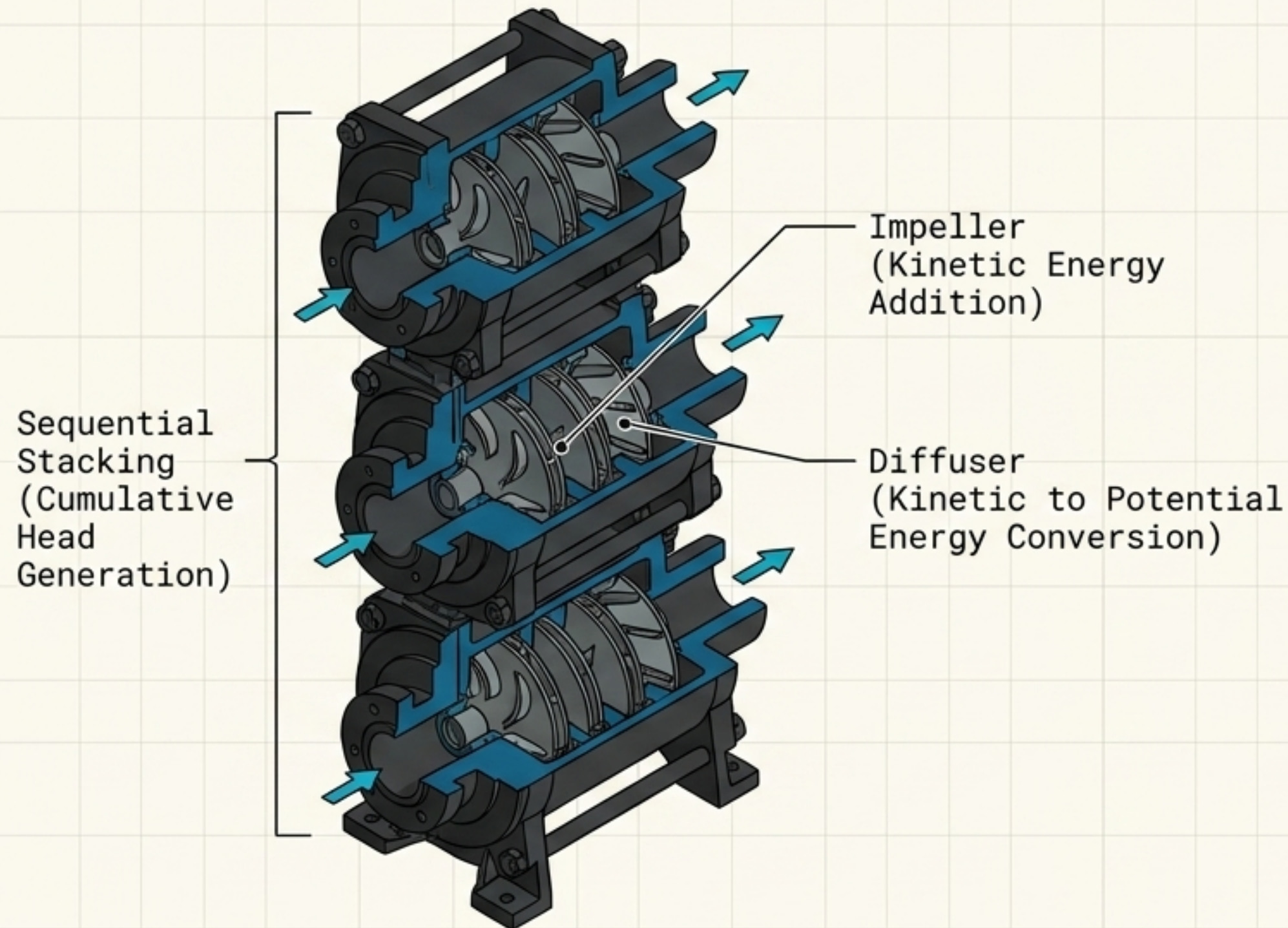
The objective is not solely to increase production, but to operate within a stable performance envelope while minimizing reliability risks and efficiency losses.



The Energy Journey: Systemic Conversion



Generating Lift: Pressure Addition & Hydraulic Head

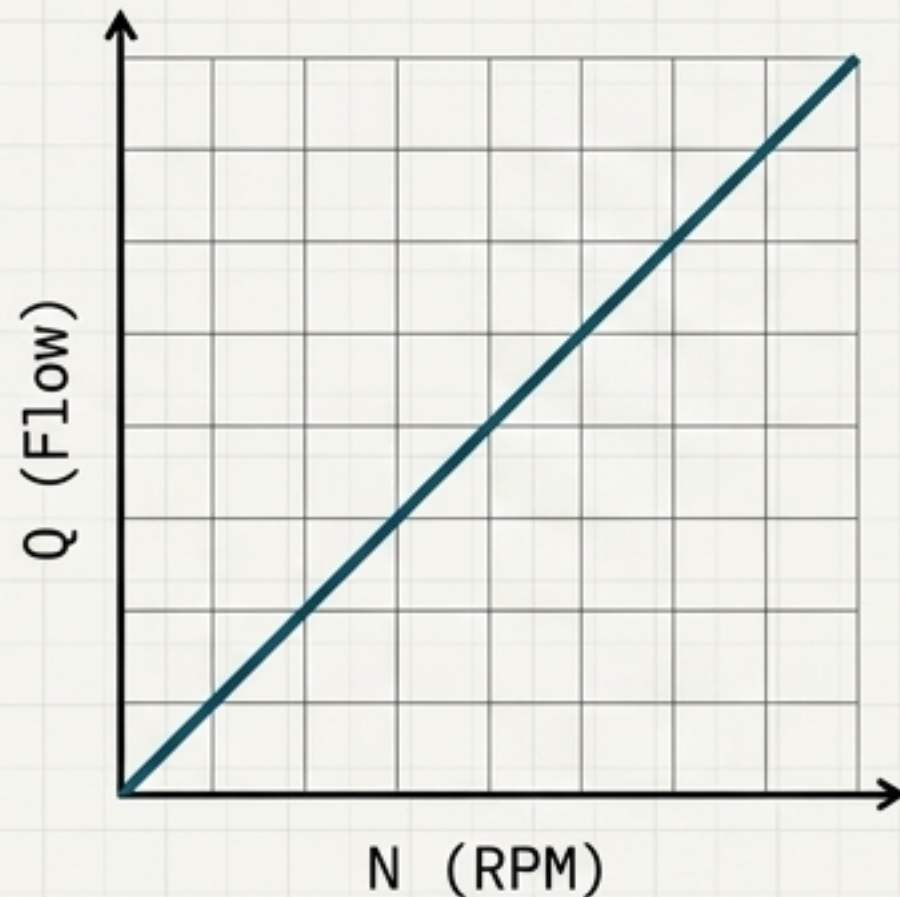


The Rules of Dynamics: Affinity Laws

Predicting system performance by varying rotational speed (RPM).

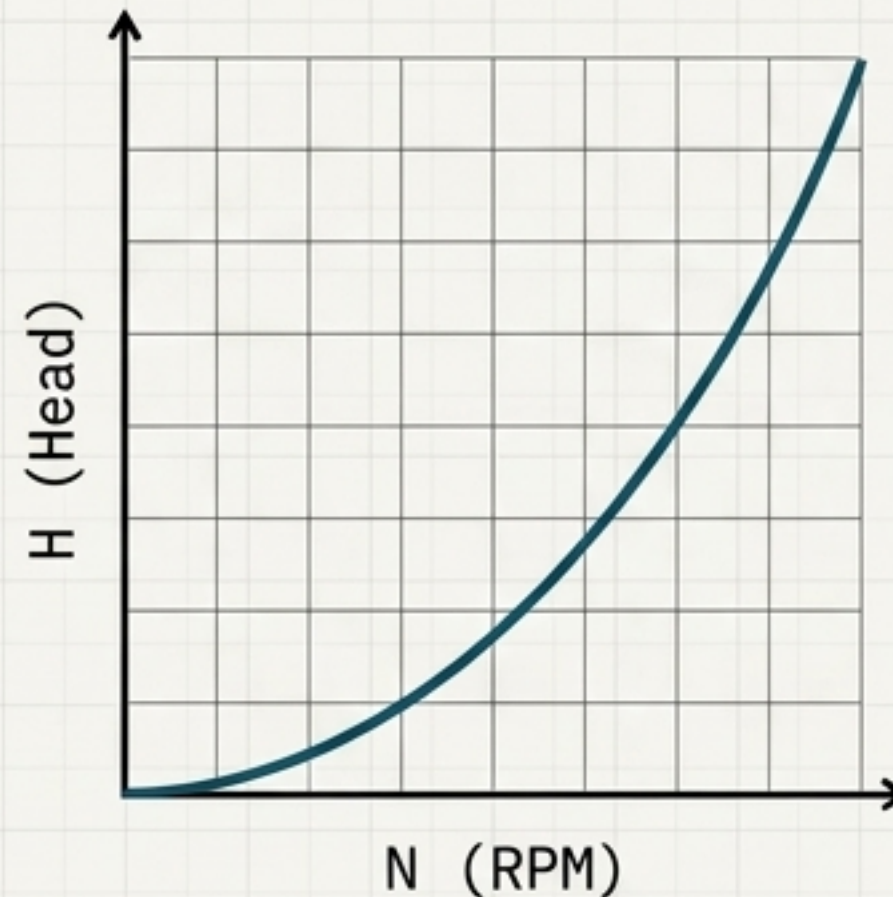
Flow (Q)

Linear Relationship
 $Q_1/Q_2 = N_1/N_2$



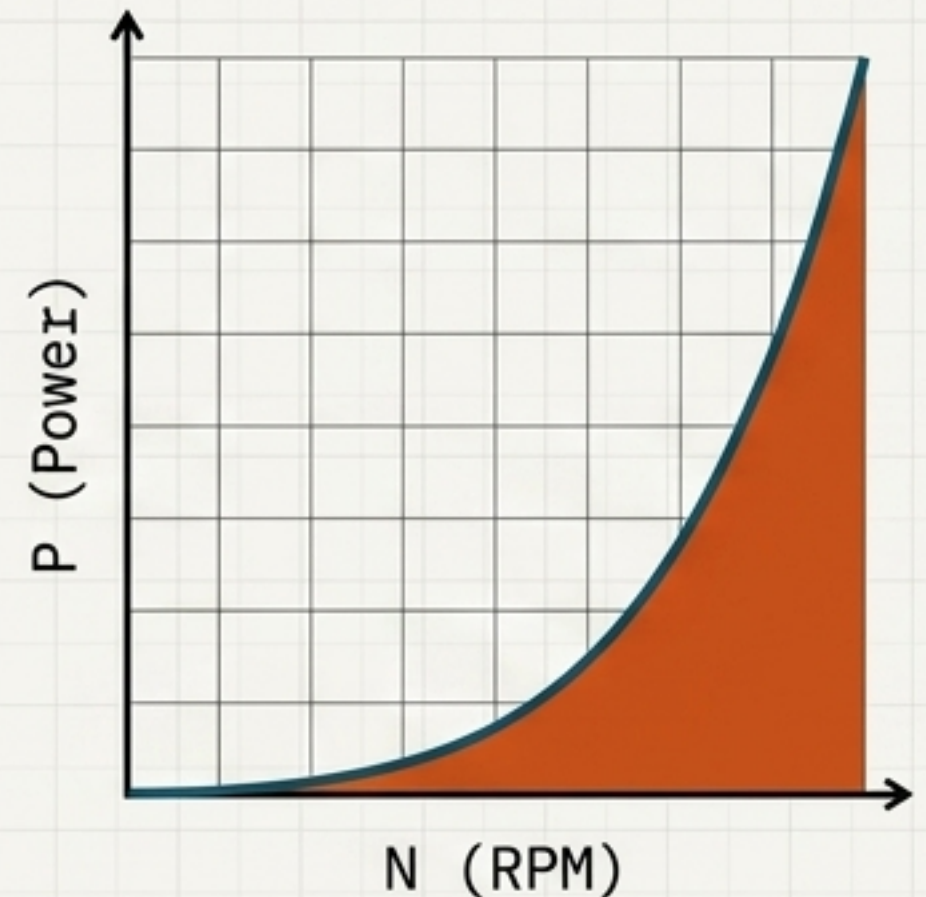
Head (H)

Square Relationship
 $H_1/H_2 = (N_1/N_2)^2$



Power (P)

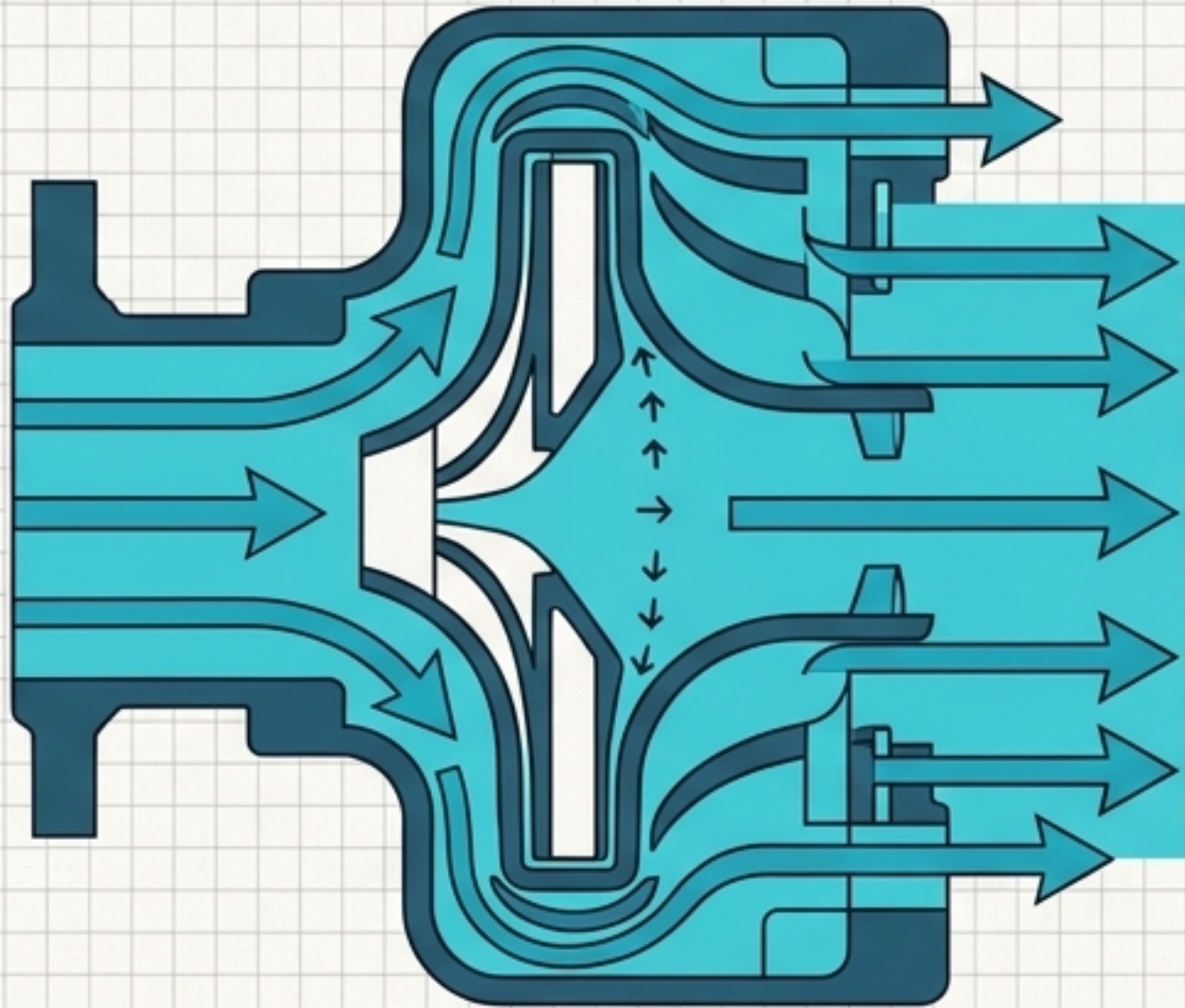
Cubic Relationship
 $P_1/P_2 = (N_1/N_2)^3$



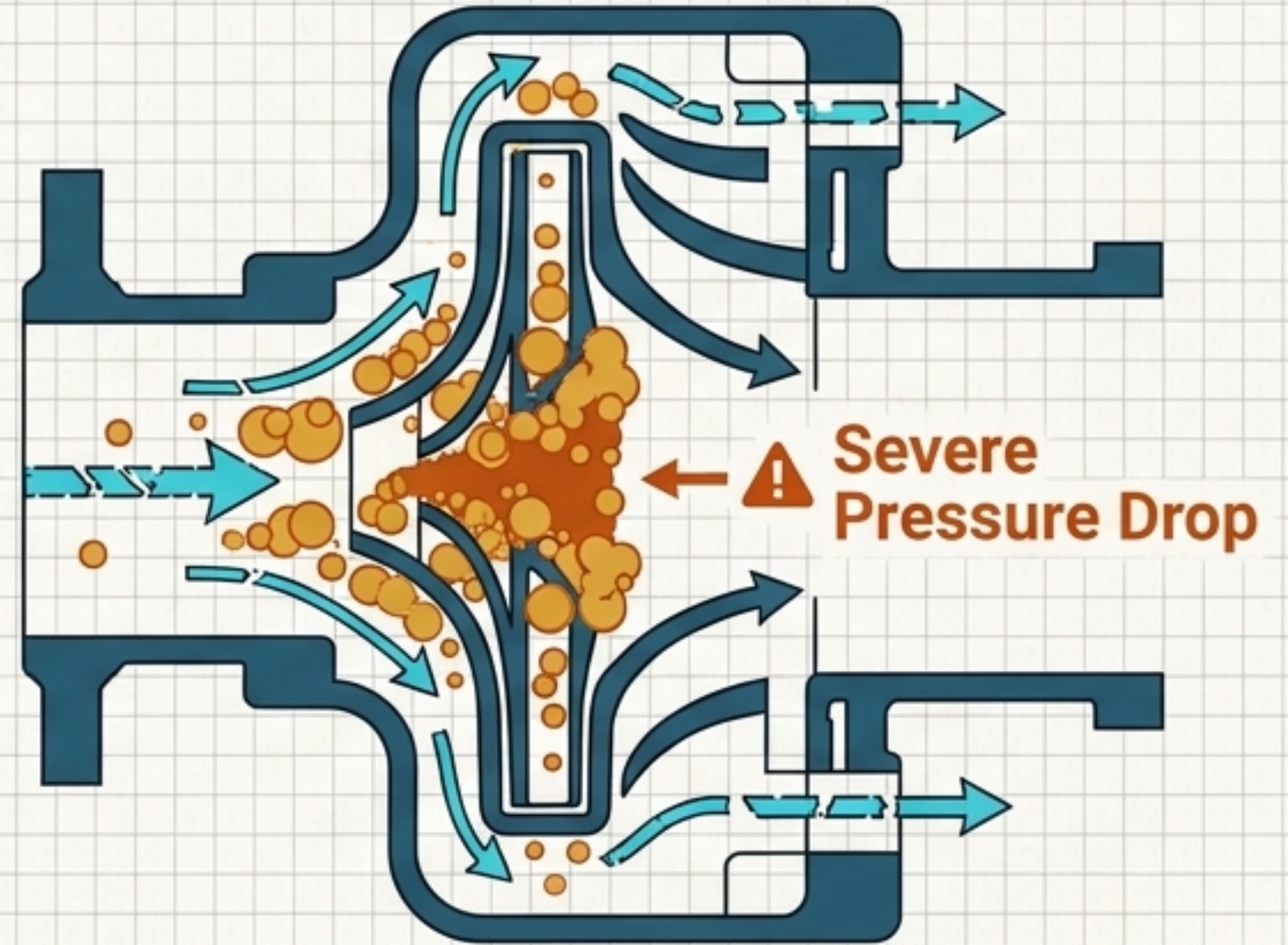
The Compressibility Threat: Gas Interference & Gas Lock

ESPs are designed for incompressible liquids. Free gas accumulates in the impeller eye, blocking fluid flow, degrading hydraulic head, and eventually halting production.

Optimal Liquid Condition



Gas Lock Condition

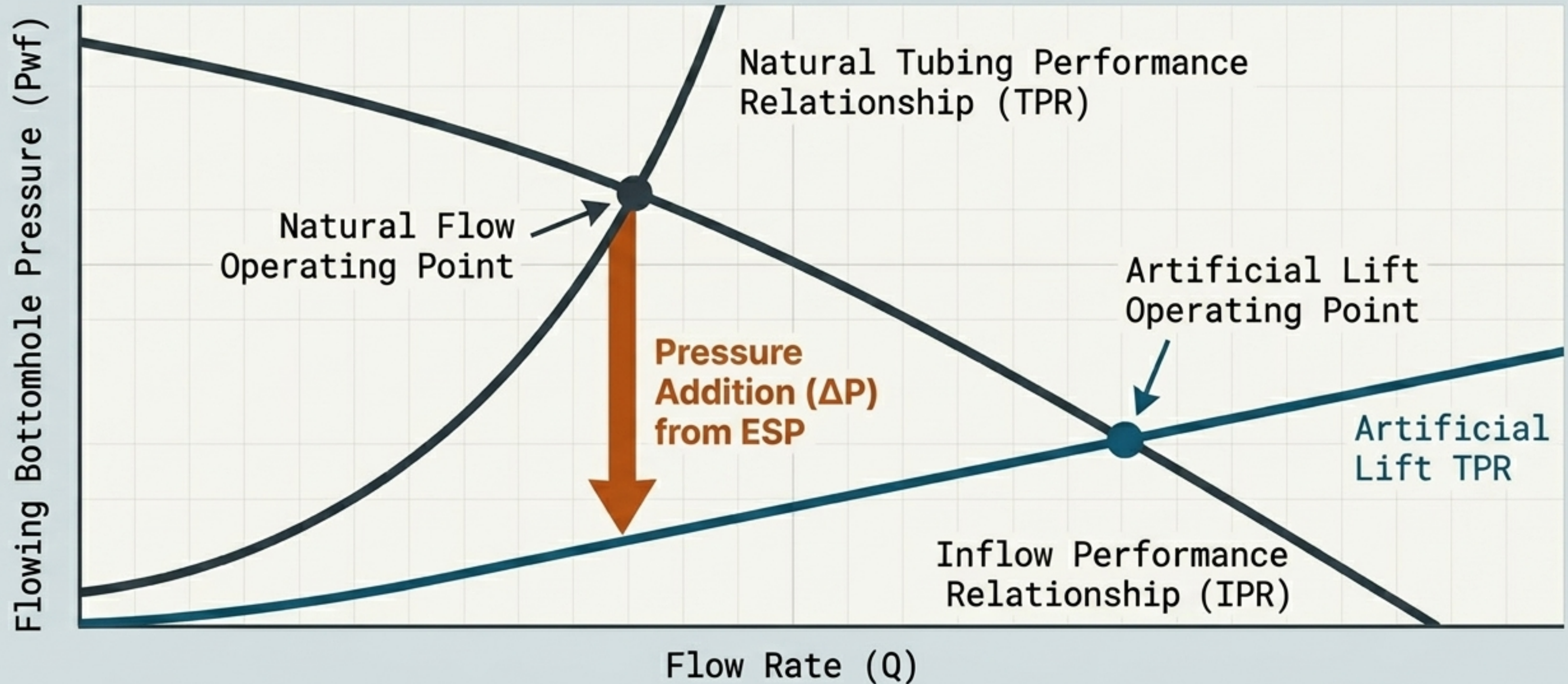


System Vulnerabilities: Common Failure Mechanisms

Failure Category	Primary Symptom	Root Cause Trigger
Mechanical Wear	Vibration, broken shafts	Abrasives (sand), operating off-curve
Electrical Degradation	Insulation failure, short circuits	Thermal cycling, voltage spikes, fluid ingress
Thermal Overload	Motor burn-out	Low fluid velocity (poor cooling), high viscosity
Hydraulic Failure	Extreme underload / fluctuating amps	Severe gas interference, pump-off conditions

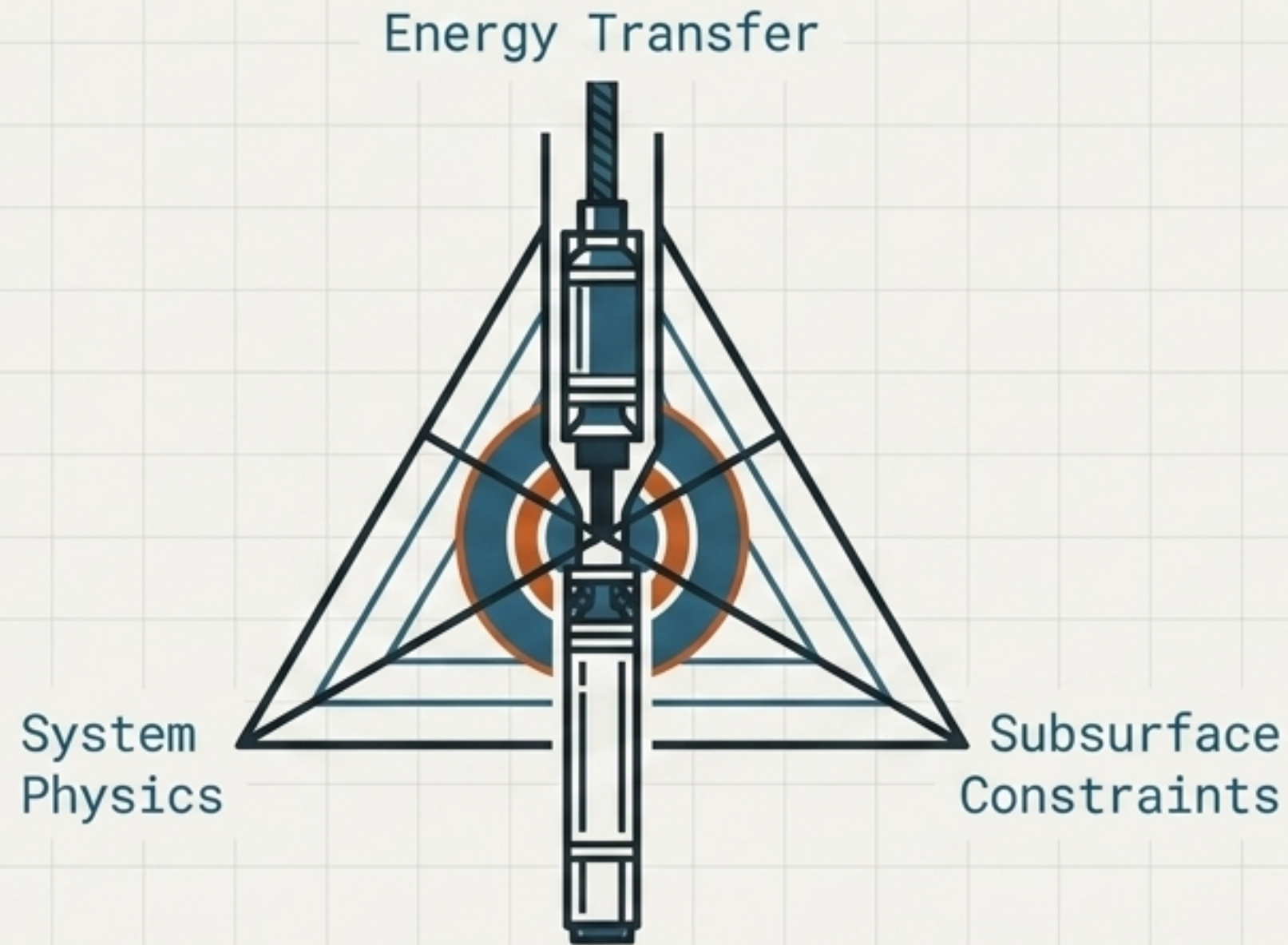
System Synthesis: Nodal-Analysis & Optimization

How the ESP modifies the well's operating point by injecting hydraulic head.



The Engineering Balance

Achieving reliable and efficient production requires mastering the interplay of energy transfer, system physics, and subsurface constraints.



Synthesized by Engineer Sajjad K. Hano.